

Matplotlib

matplotlib is a python library it is used for data visualisation

importing matplotlib

```
In [1]: import matplotlib.pyplot as plt  
import numpy as np
```

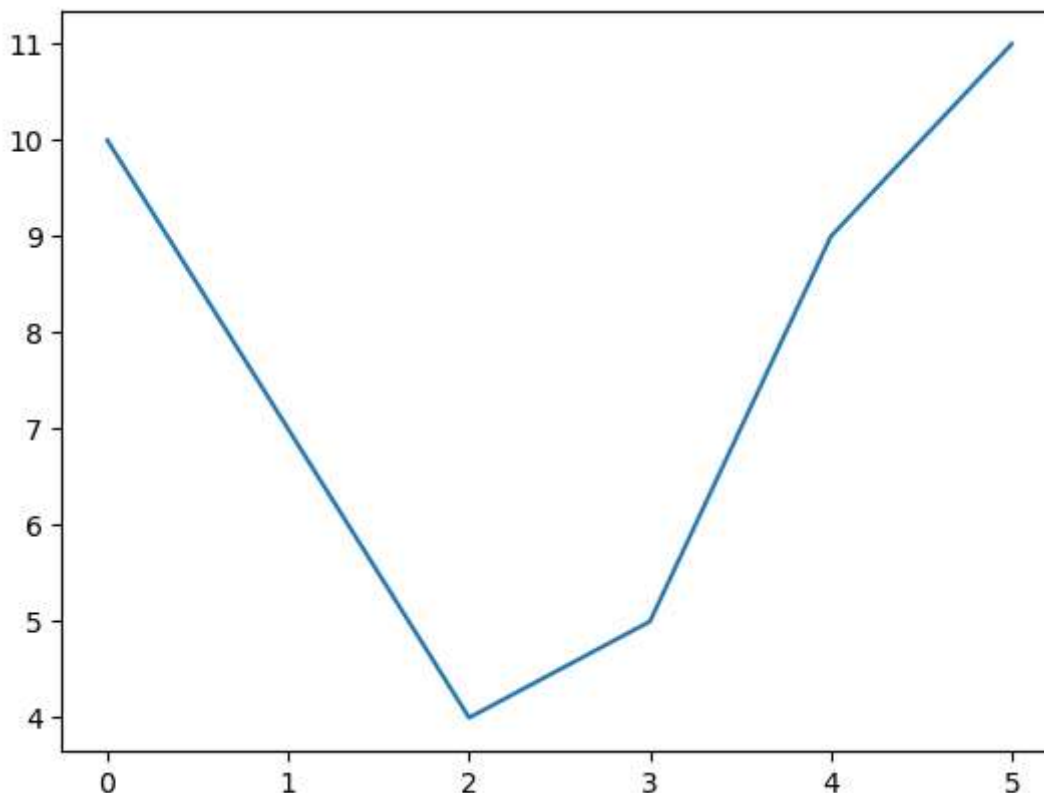
Essential Role of Pyplot in Matplotlib

pyplot is a module in matplotlib it provide functions like plot(),bar(),pie(),hist(),scatter() functions helps us to make plots easily

Line Graph

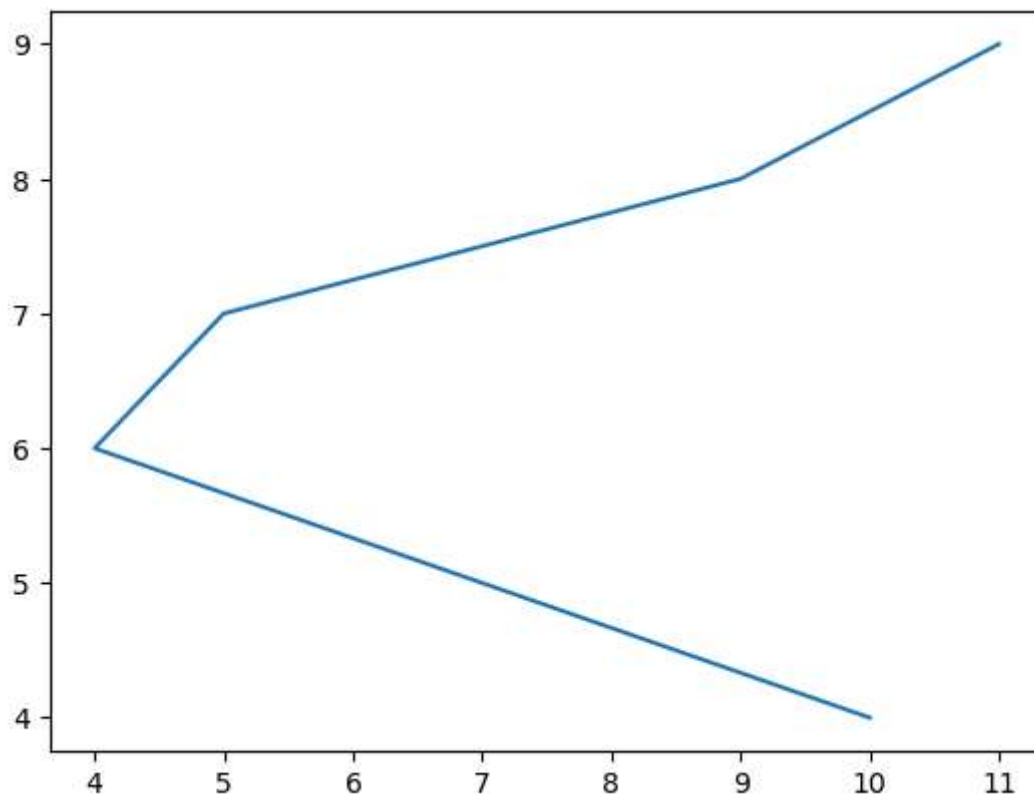
Line graph is used when you want to show continuous data and how it is changing over trends

```
In [2]: x=np.array([10,7,4,5,9,11])  
plt.plot(x)  
plt.show()
```



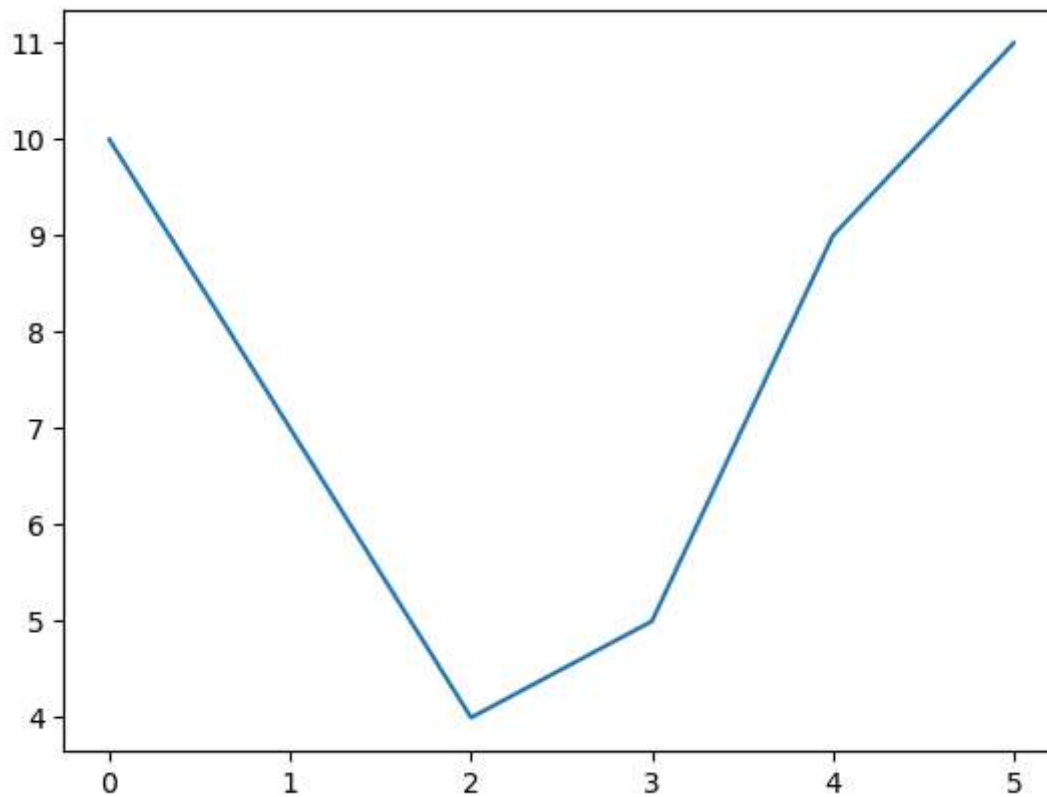
only given y-axis variables we plot the and consider x axis as index of y-axis variable

```
In [3]: x=np.array([10,7,4,5,9,11])  
y=np.array([4,5,6,7,8,9])  
plt.plot(x,y)  
plt.show()
```

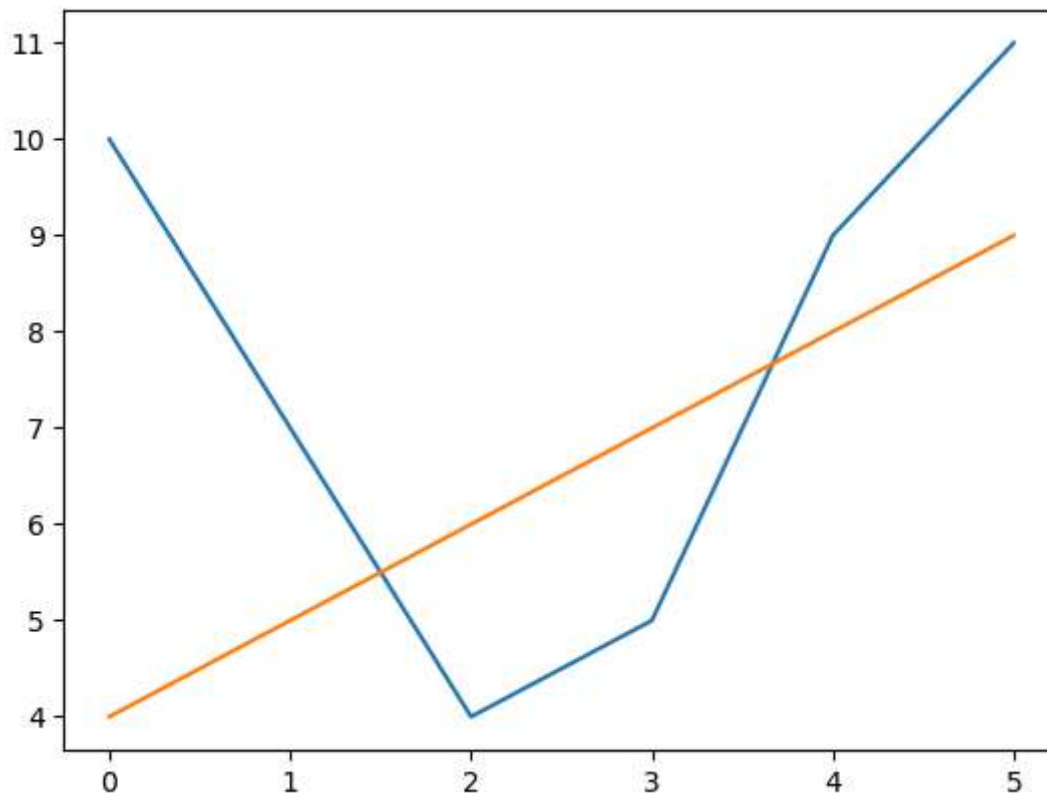


```
In [4]: x1=np.array([10,7,4,5,9,11])  
x2=np.array([4,5,6,7,8,9])  
plt.plot(x1)
```

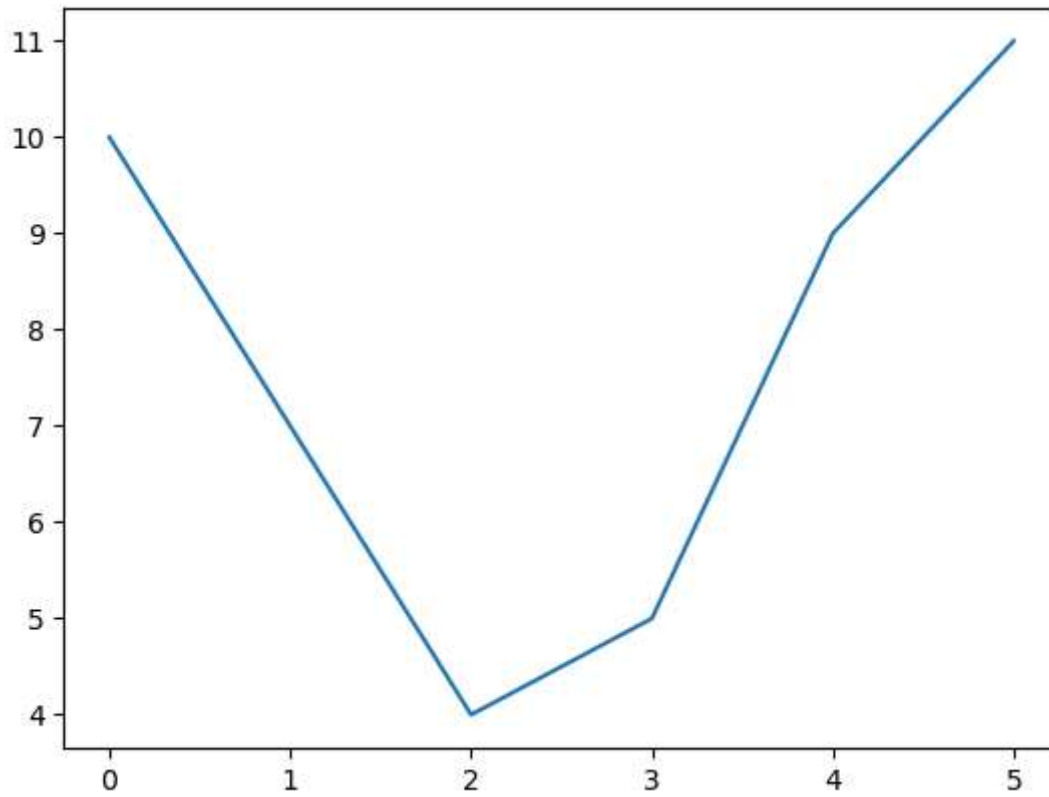
```
Out[4]: [<matplotlib.lines.Line2D at 0x13ca458f610>]
```

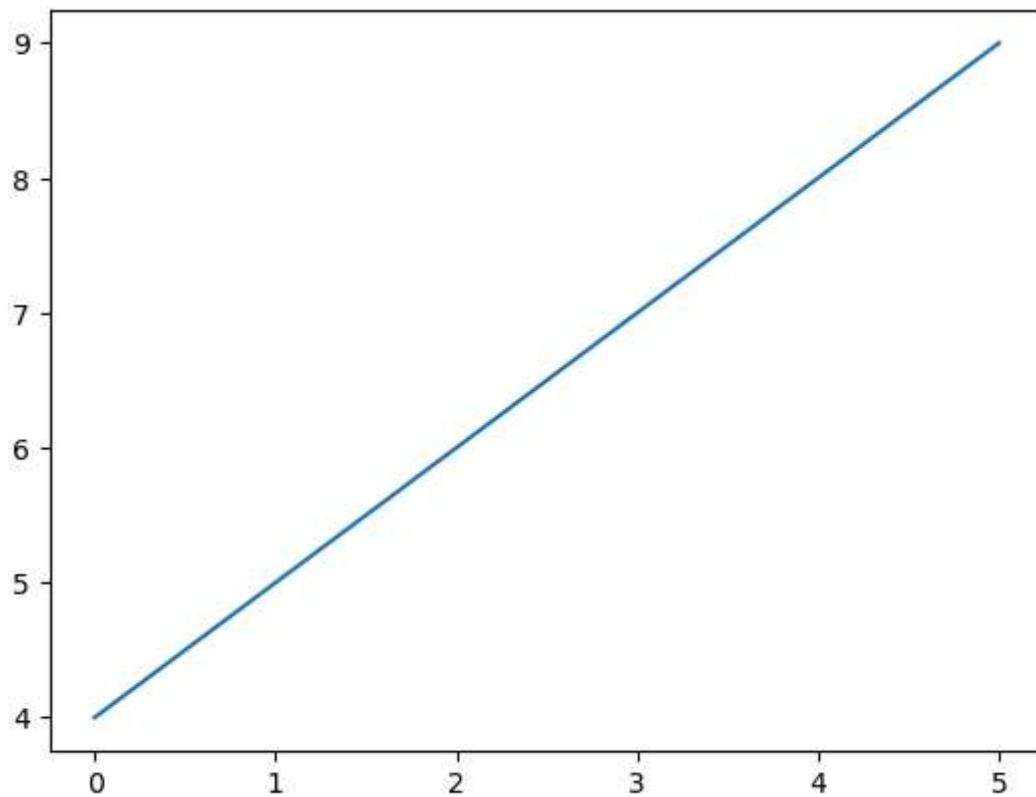


```
In [5]: x1=np.array([10,7,4,5,9,11])  
x2=np.array([4,5,6,7,8,9])  
plt.plot(x1)  
plt.plot(x2)  
plt.show()
```

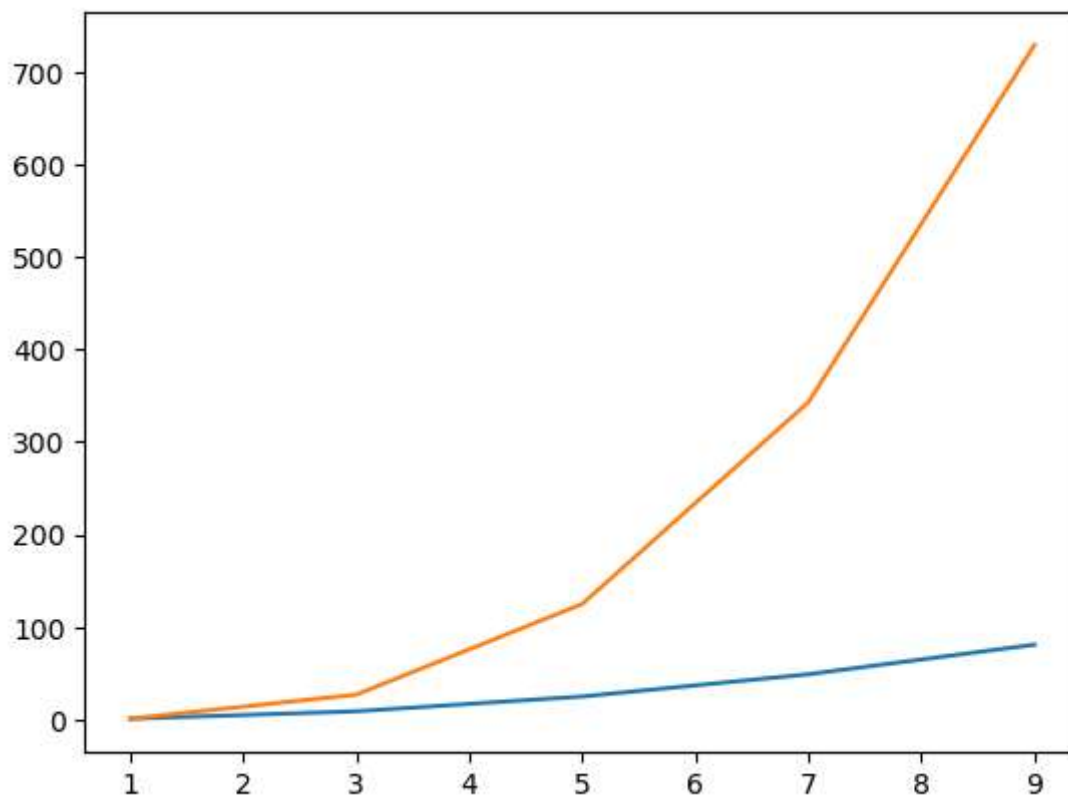


```
In [6]: x1=np.array([10,7,4,5,9,11])  
plt.plot(x1)  
plt.show()  
x2=np.array([4,5,6,7,8,9])  
plt.plot(x2)  
plt.show()
```



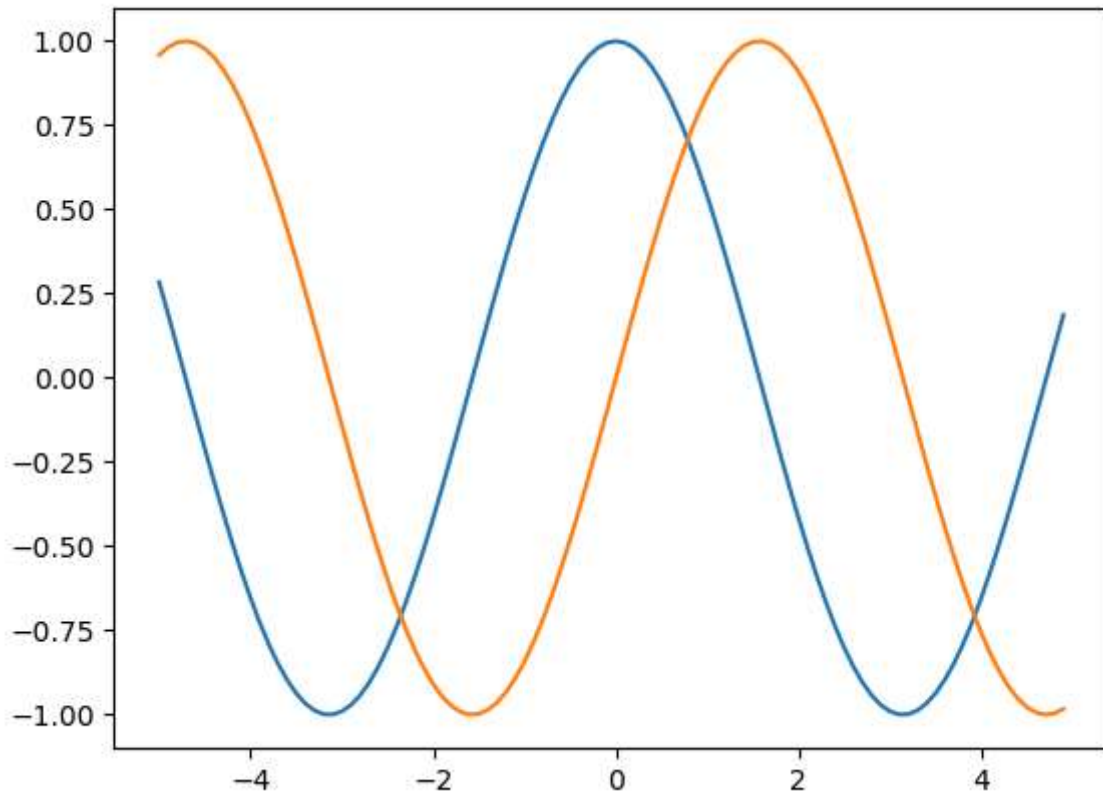


```
In [7]: x=np.array([1,3,5,7,9])  
plt.plot(x,x**2)  
plt.plot(x,x**3)  
plt.show()
```



in `plt.plot` pass arguments based on our requirement if we pass one argument it consider by default on y axis it plot x axis based on indexes

```
In [8]: x=np.arange(-5,5,0.1)
plt.plot(x,np.cos(x))
plt.plot(x,np.sin(x))
plt.show()
```

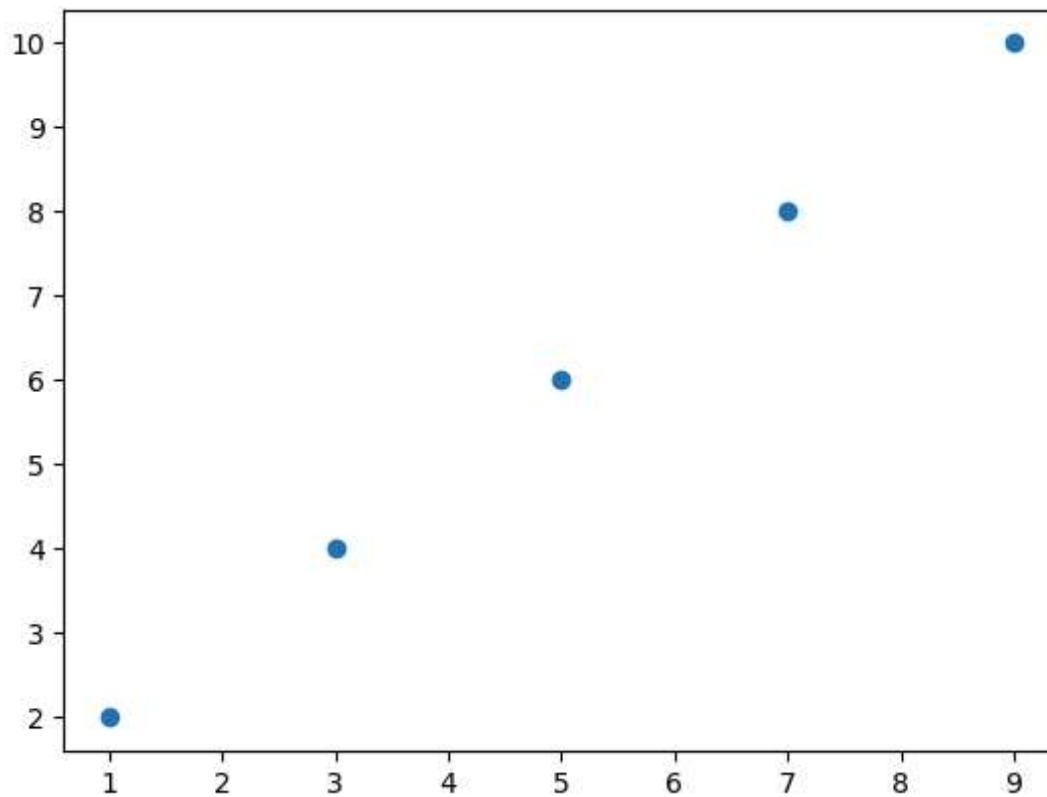


scatter plot

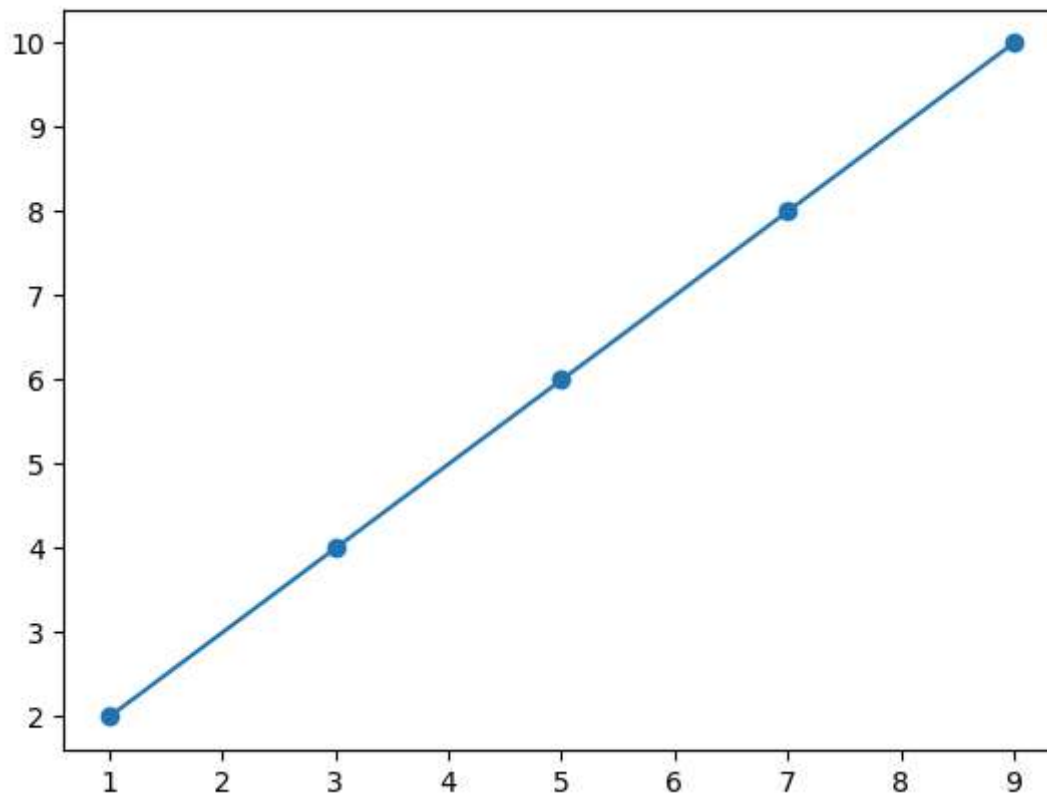
scatter plot is used for visualizing relation between two continuous variables and it is used to find outliers, correlation

in scatter plot we have to give both x and y arguments

```
In [9]: x=np.array([1,3,5,7,9])
y=np.array([2,4,6,8,10])
plt.scatter(x,y)
plt.show()
```

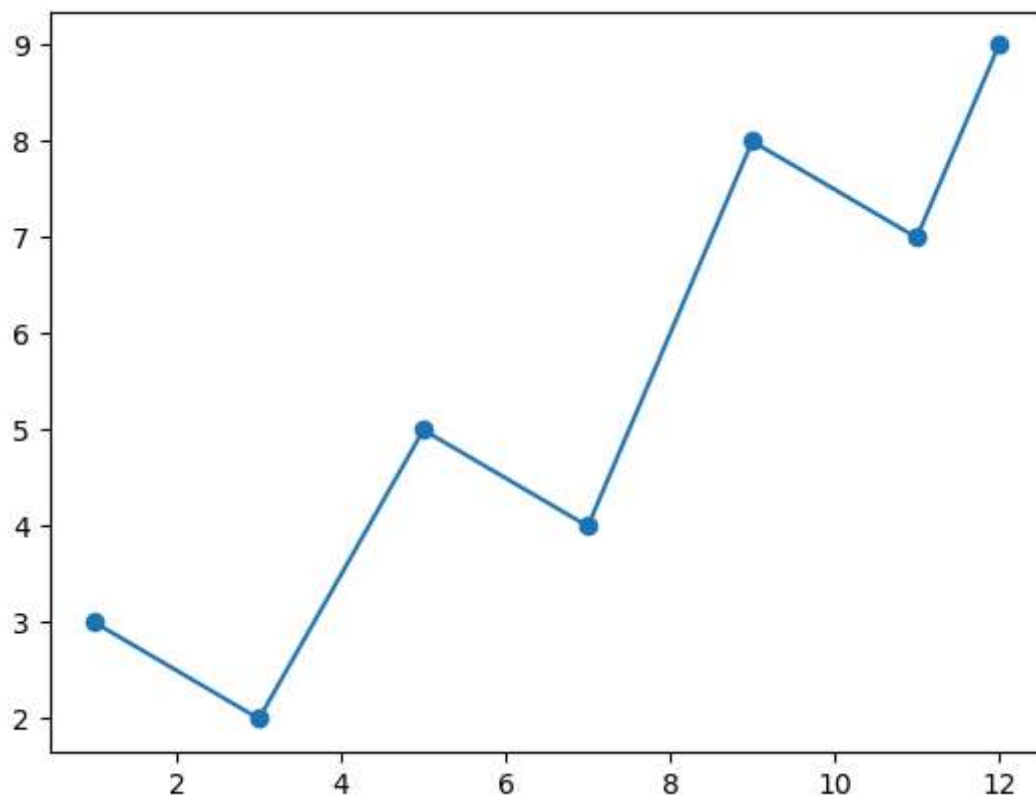


```
In [10]: x=np.array([1,3,5,7,9])  
y=np.array([2,4,6,8,10])  
plt.scatter(x,y)  
plt.plot(x,y)  
plt.show()
```

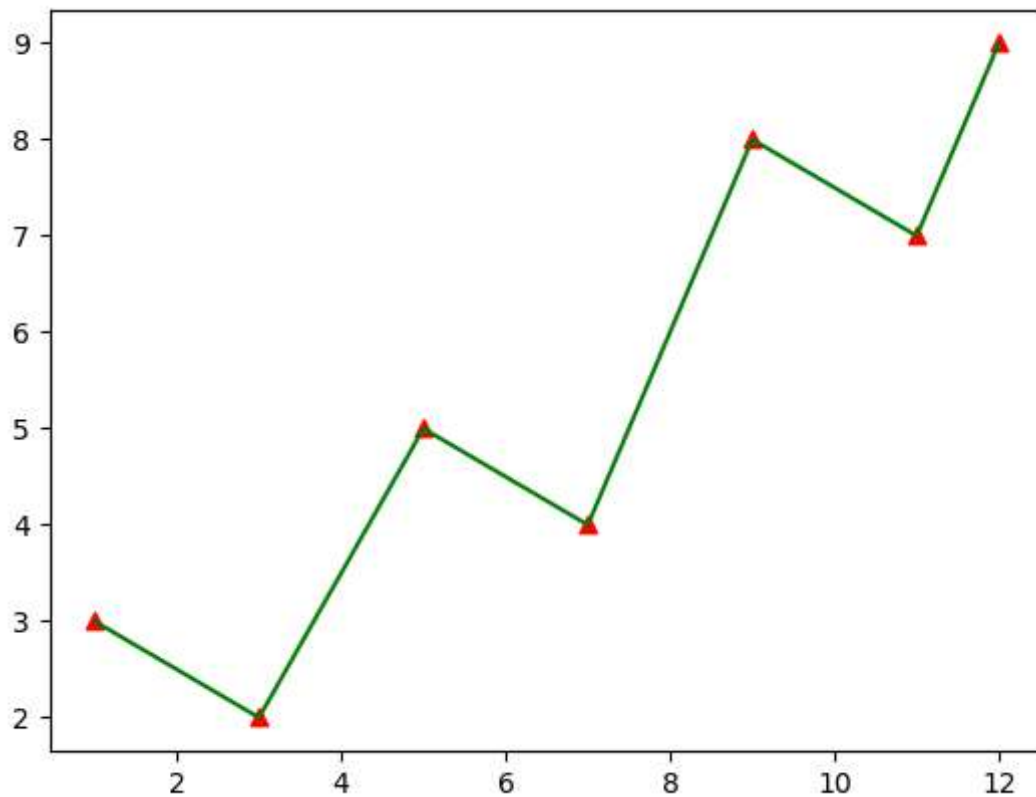


we can use both `scatter` and `plt.plot` function to get this kind of graph

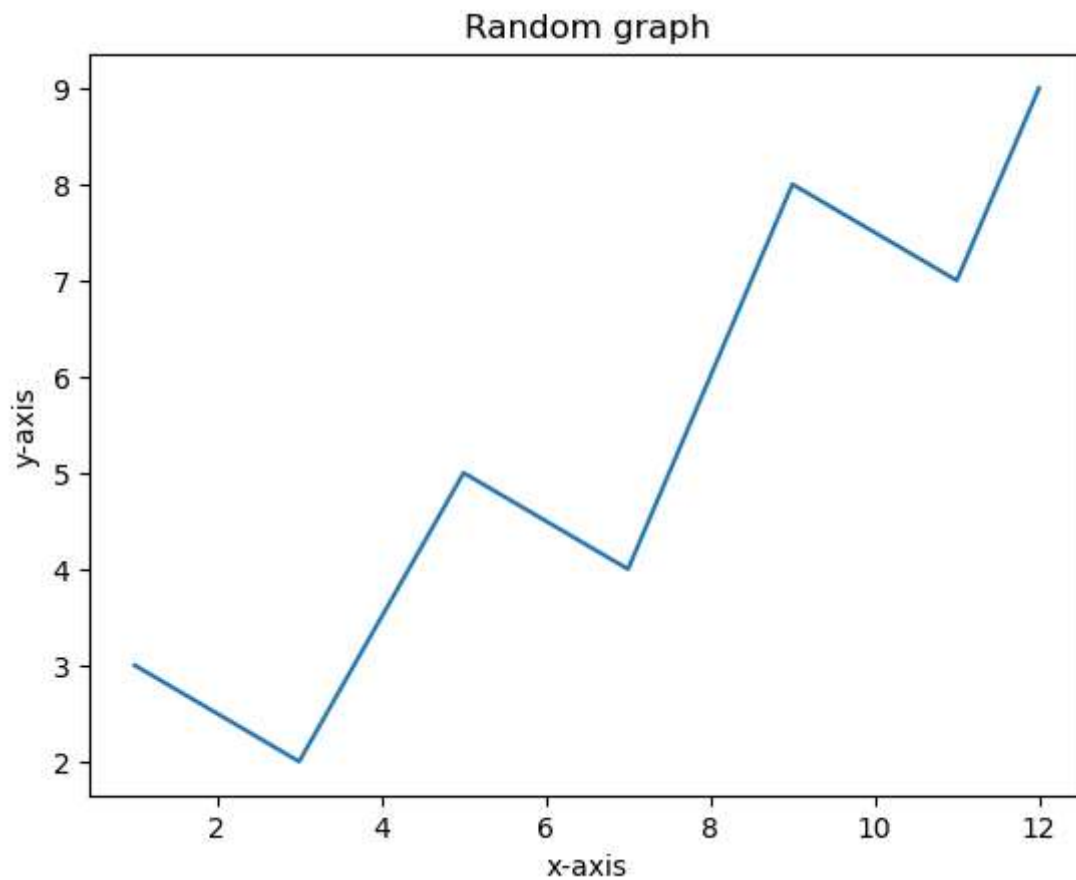
```
In [11]: x=np.array([1,3,5,7,9,11,12])
y=np.array([3,2,5,4,8,7,9])
plt.scatter(x,y)
plt.plot(x,y)
plt.show()
```



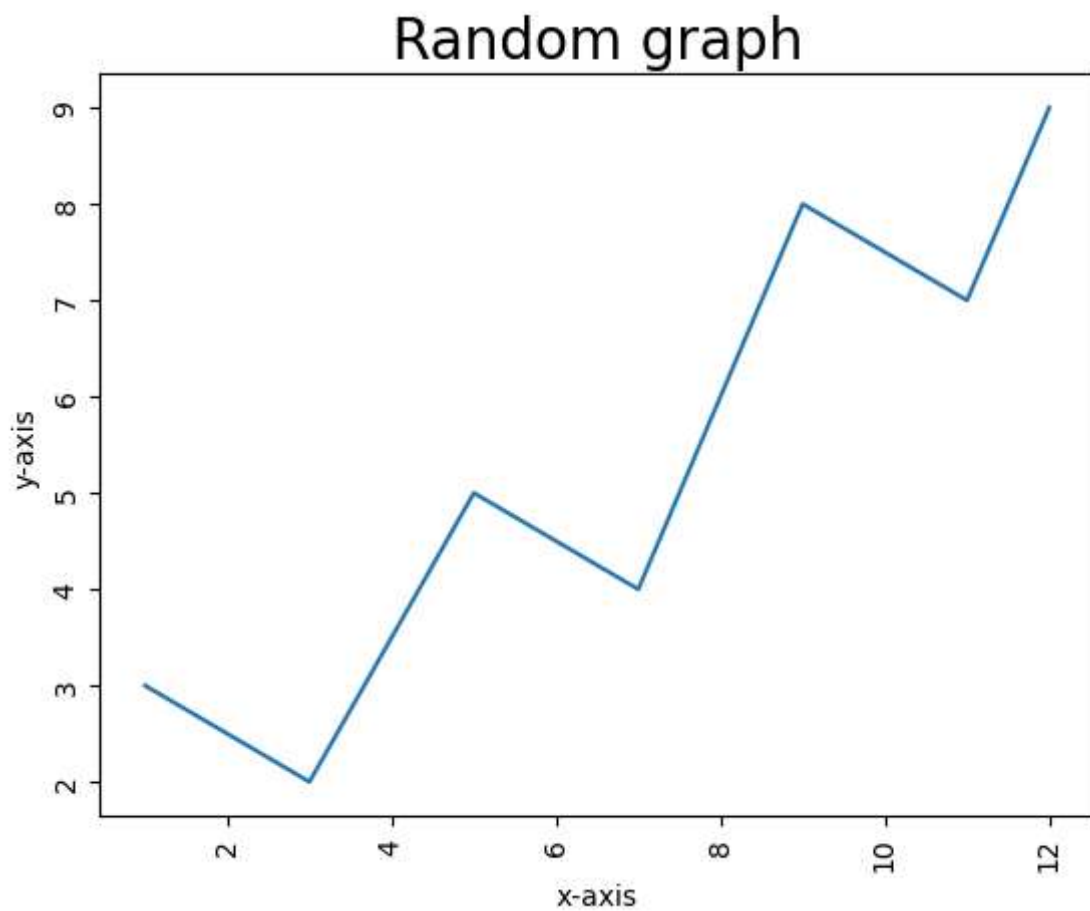
```
In [12]: x=np.array([1,3,5,7,9,11,12])
y=np.array([3,2,5,4,8,7,9])
plt.scatter(x,y,color='red',marker='^')
plt.plot(x,y,color='green')
plt.show()
```

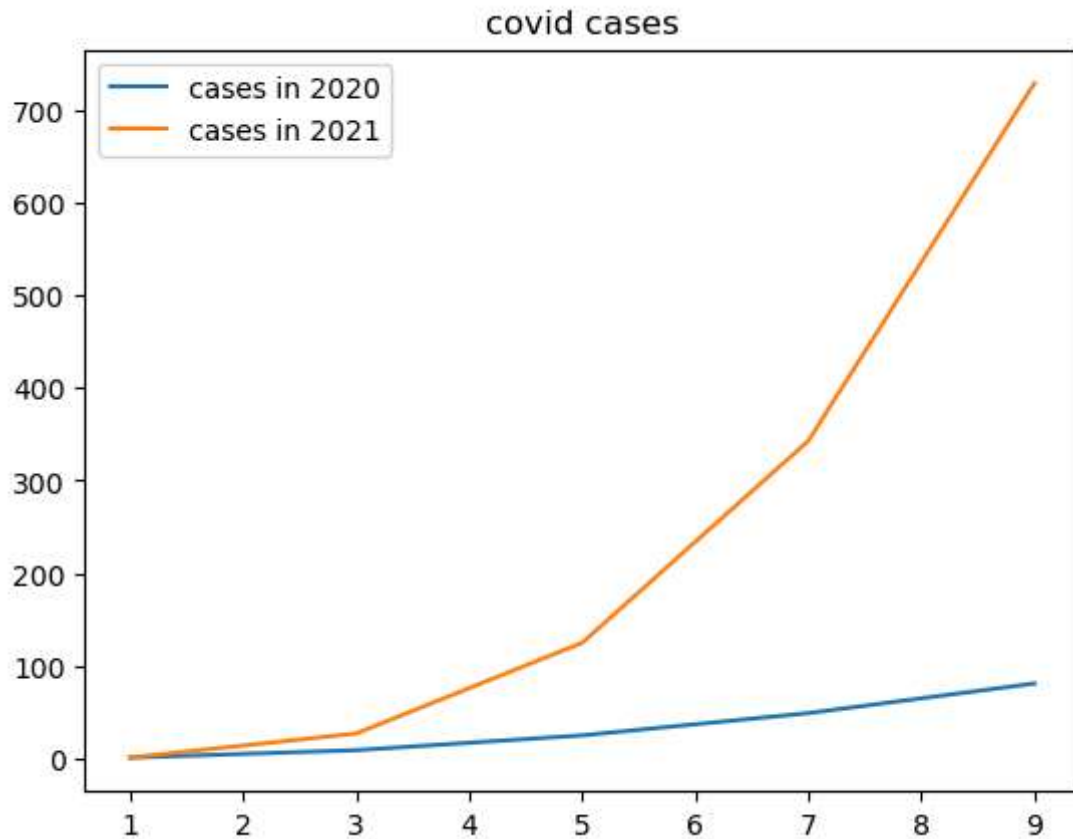
```
In [13]: x=np.array([1,3,5,7,9,11,12])
y=np.array([3,2,5,4,8,7,9])
plt.title("Random graph")
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.plot(x,y)
plt.show()
```



```
In [14]: x=np.array([1,3,5,7,9,11,12])
y=np.array([3,2,5,4,8,7,9])
plt.title("Random graph",fontsize=20)
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.xticks(rotation=90)
plt.yticks(rotation=90)
plt.plot(x,y)
plt.show()
```

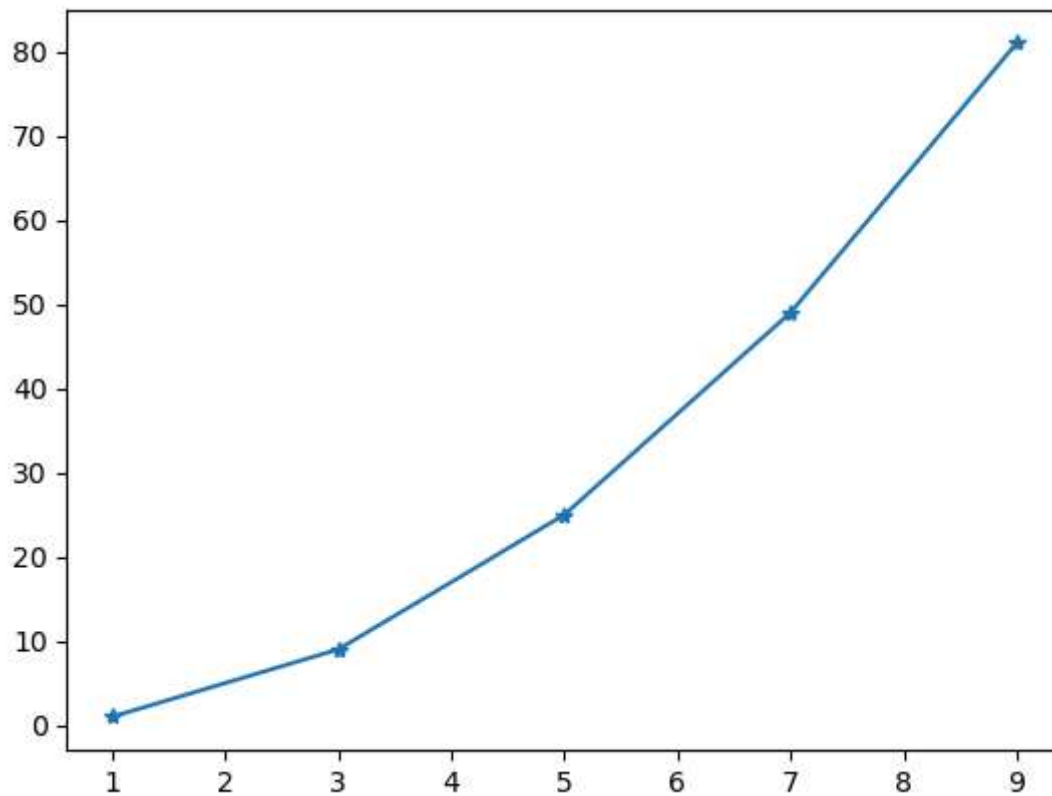


```
In [15]: x=np.array([1,3,5,7,9])
plt.plot(x,x**2,label="cases in 2020")
plt.plot(x,x**3,label="cases in 2021")
plt.title("covid cases")
plt.legend()
plt.show()
```



```
In [16]: x=np.array([1,3,5,7,9])
plt.plot(x,x**2,label="cases in 2020",marker="*")

plt.show()
```

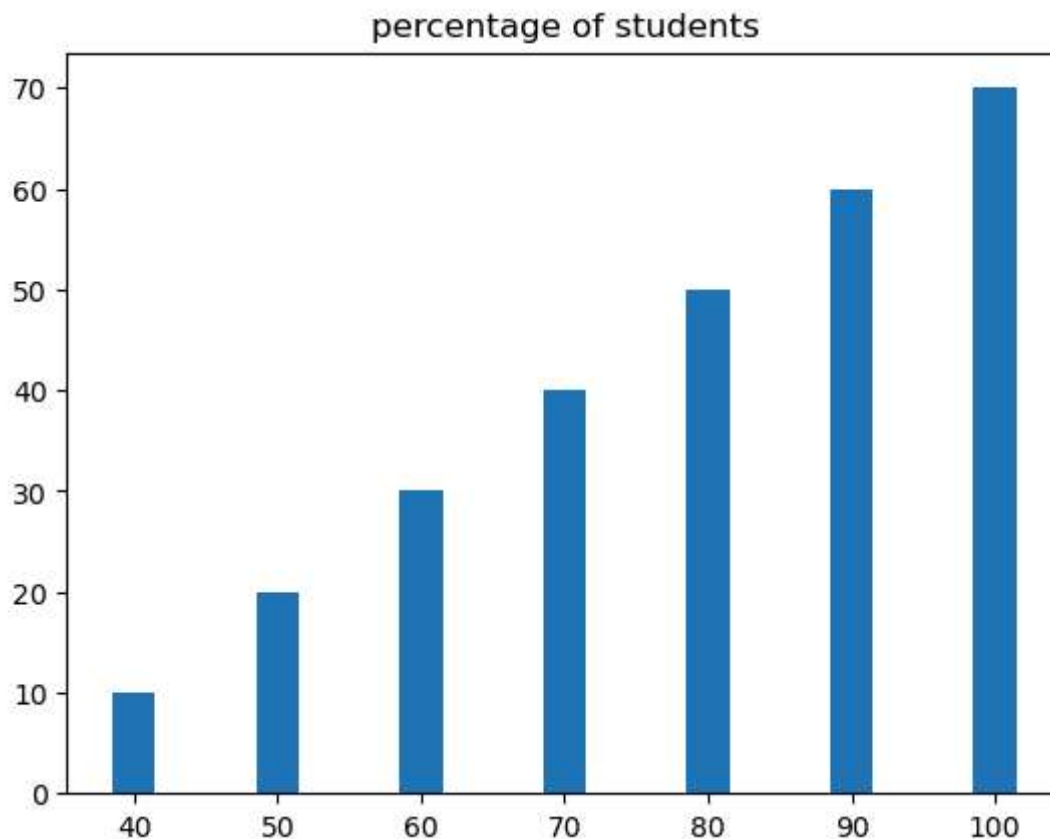


bar plot

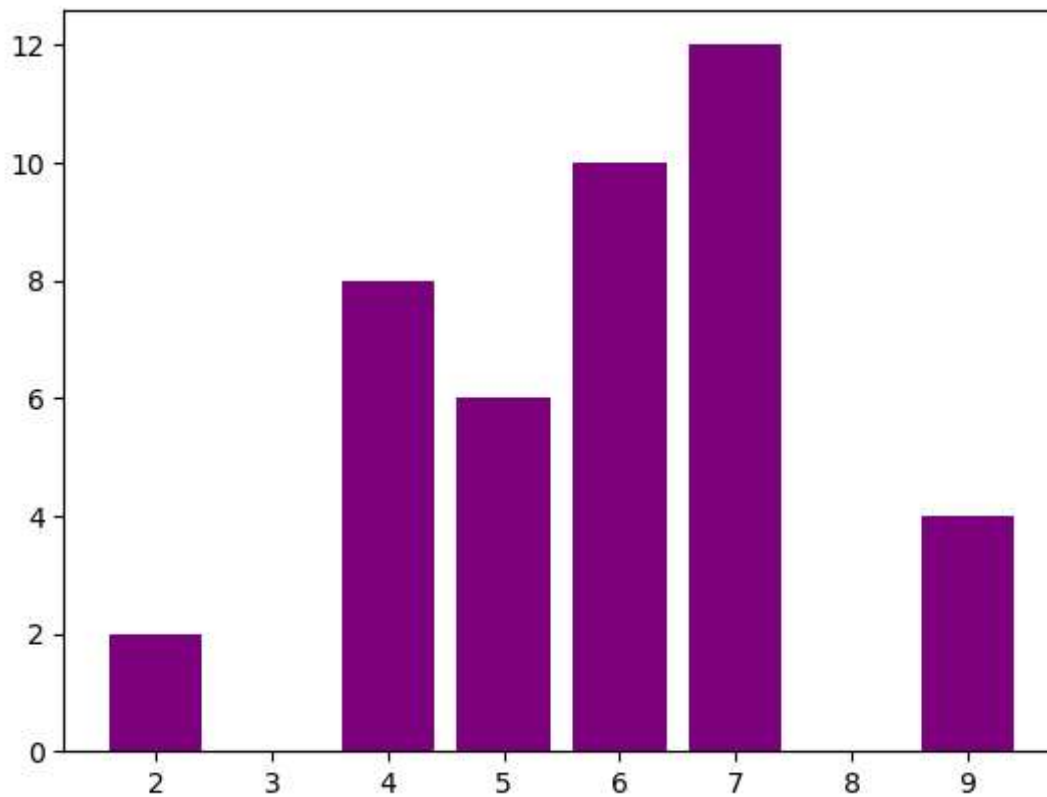
bar chart is mainly used for categorical data and for comparing different categories

bar chart is a graph that uses rectangular bars to represent data

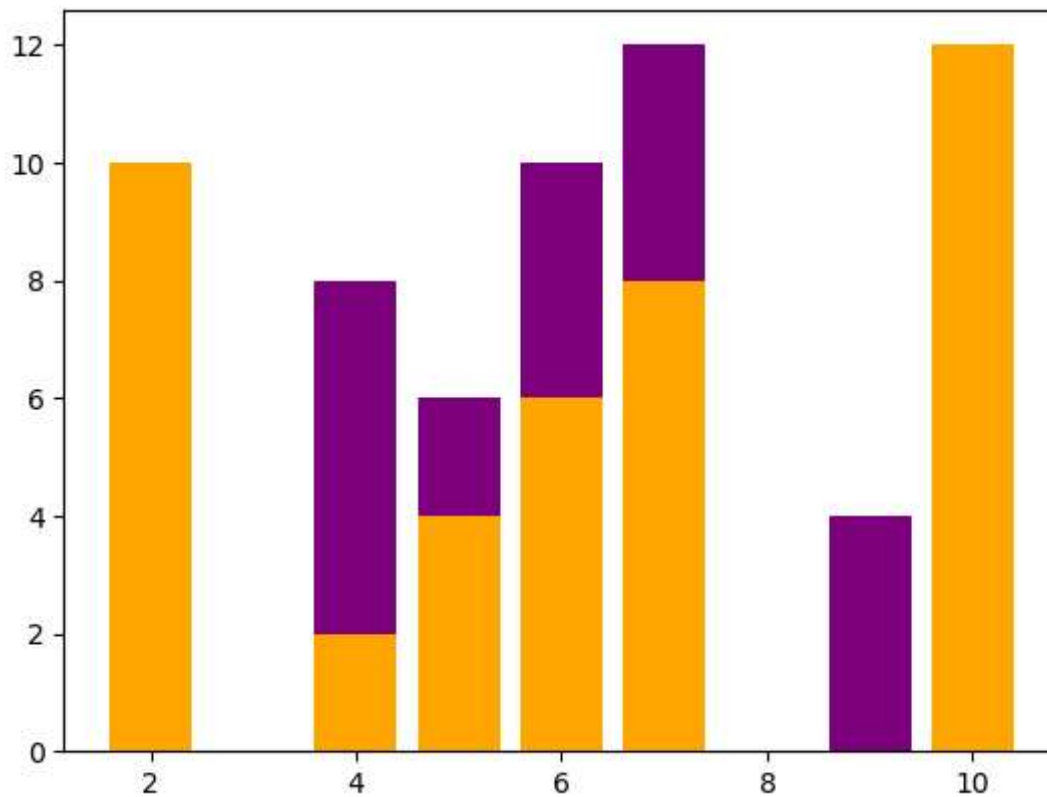
```
In [17]: percent_marks=np.array([40,50,60,70,80,90,100])
num_students=np.array([10,20,30,40,50,60,70])
plt.title("percentage of students")
plt.bar(percent_marks,num_students,width=3)
plt.show()
```



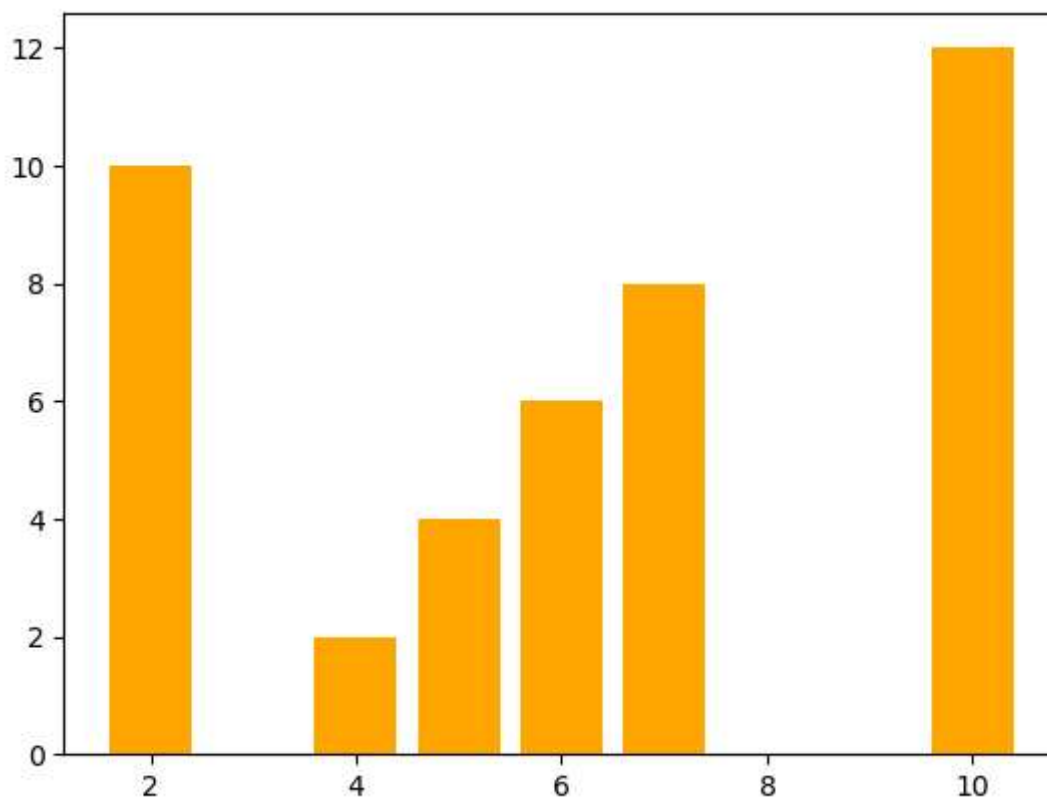
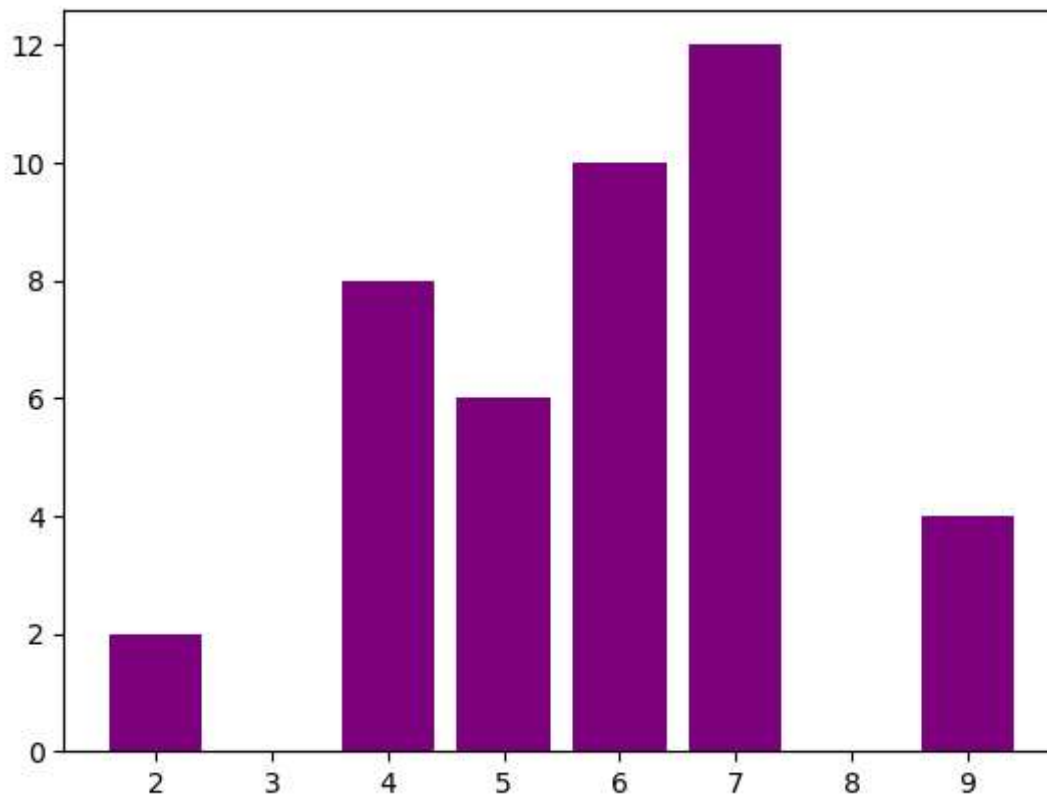
```
In [18]: x1=np.array([2,9,5,4,6,7])
y1=np.array([2,4,6,8,10,12])
plt.bar(x1,y1,color="purple")
plt.show()
```



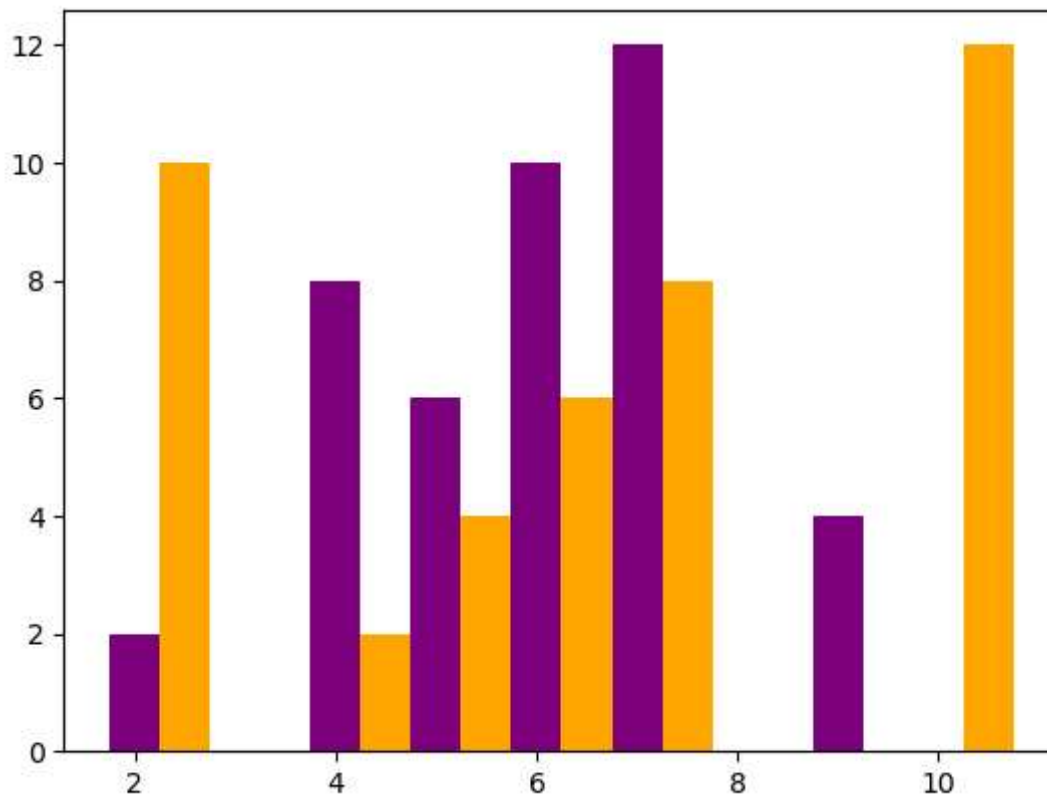
```
In [19]: x1=np.array([2,9,5,4,6,7])
          y1=np.array([2,4,6,8,10,12])
          plt.bar(x1,y1,color="purple")
          x2=np.array([4,5,6,7,2,10])
          plt.bar(x2,y1,color='orange')
          plt.show()
```



```
In [20]: x1=np.array([2,9,5,4,6,7])
y1=np.array([2,4,6,8,10,12])
plt.bar(x1,y1,color="purple")
plt.show()
x2=np.array([4,5,6,7,2,10])
plt.bar(x2,y1,color='orange')
plt.show()
```

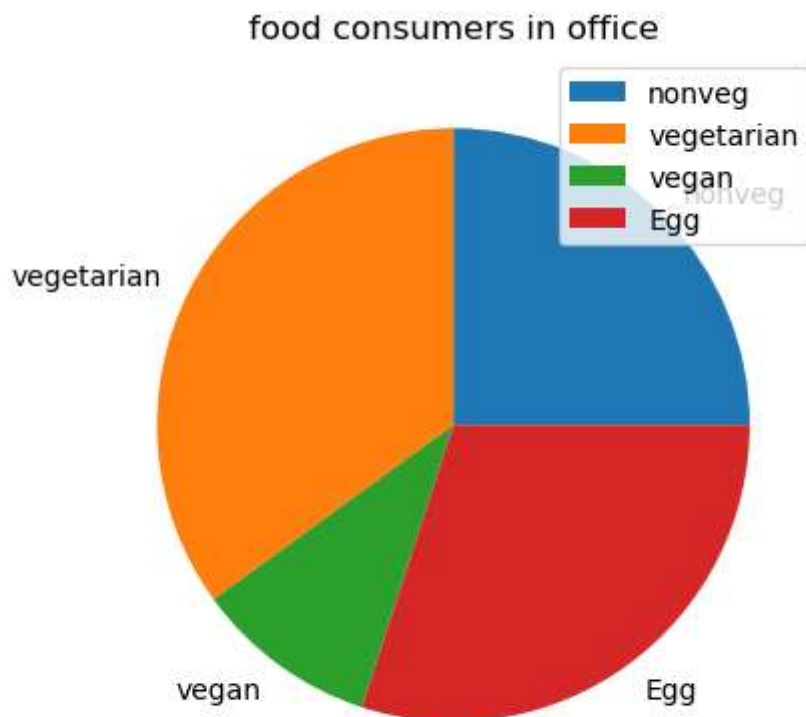


```
In [21]: x1=np.array([2,9,5,4,6,7])
y1=np.array([2,4,6,8,10,12])
plt.bar(x1,y1,color="purple",width=0.5)
x2=np.array([4,5,6,7,2,10])
plt.bar(x2+0.5,y1,width=0.5,color='orange')
plt.show()
```

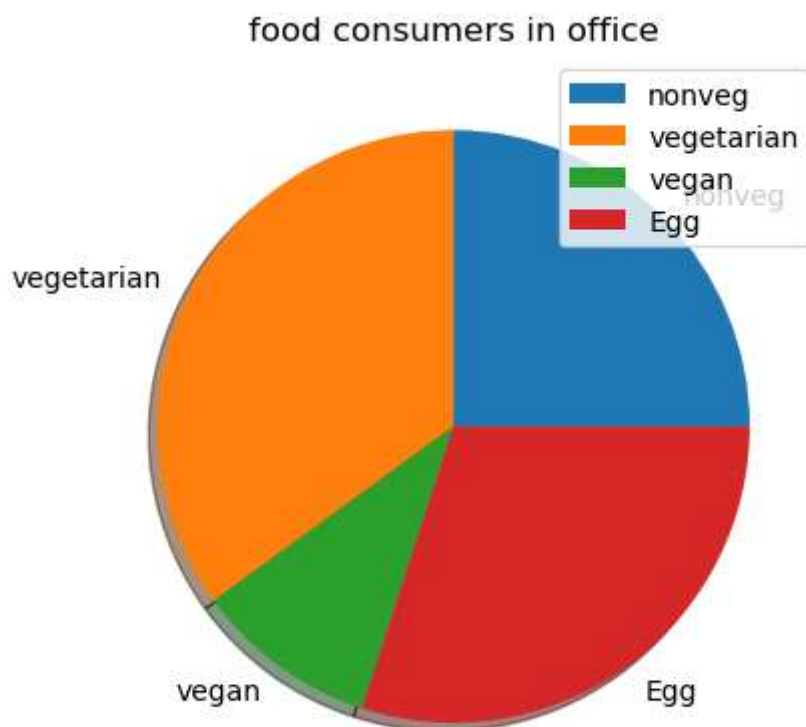



pie graph

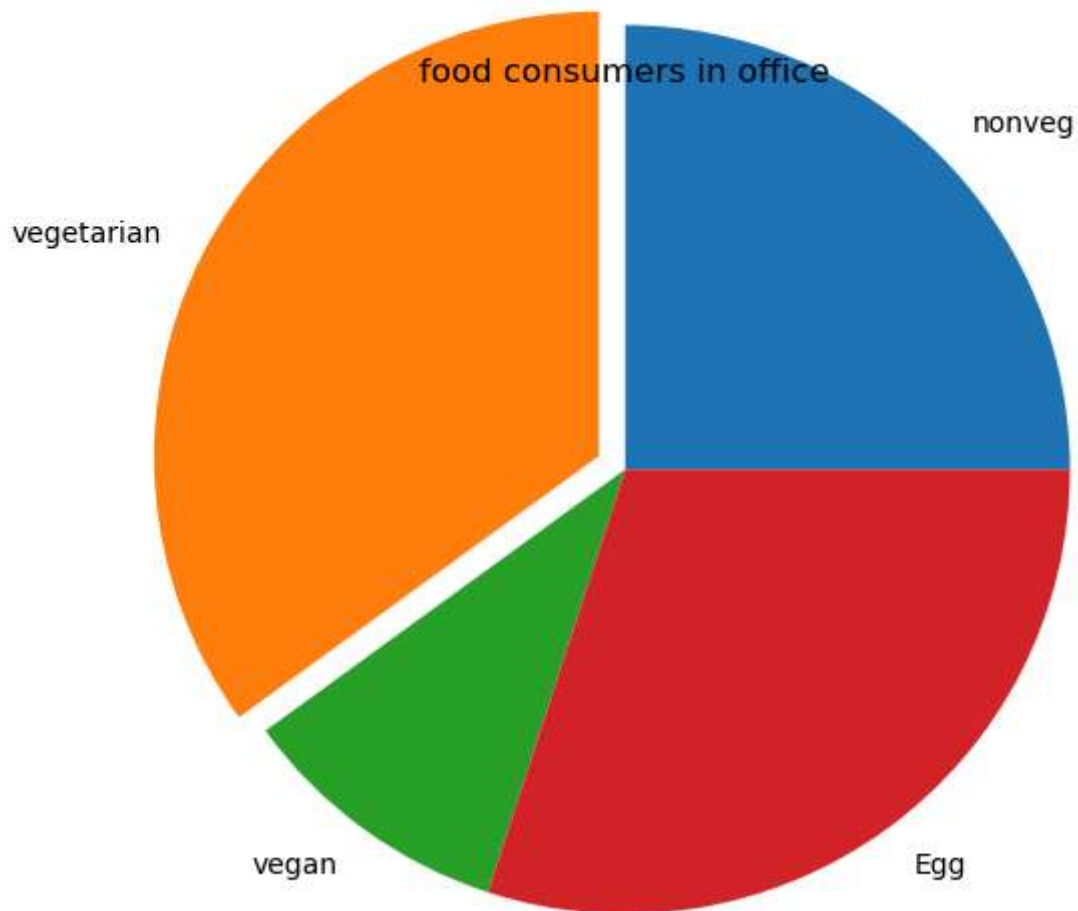
```
In [22]: values=[25,35,10,30]
label=["nonveg","vegetarian","vegan","Egg"]
plt.pie(values,labels=label)
plt.legend()
plt.title("food consumers in office")
plt.show()
```



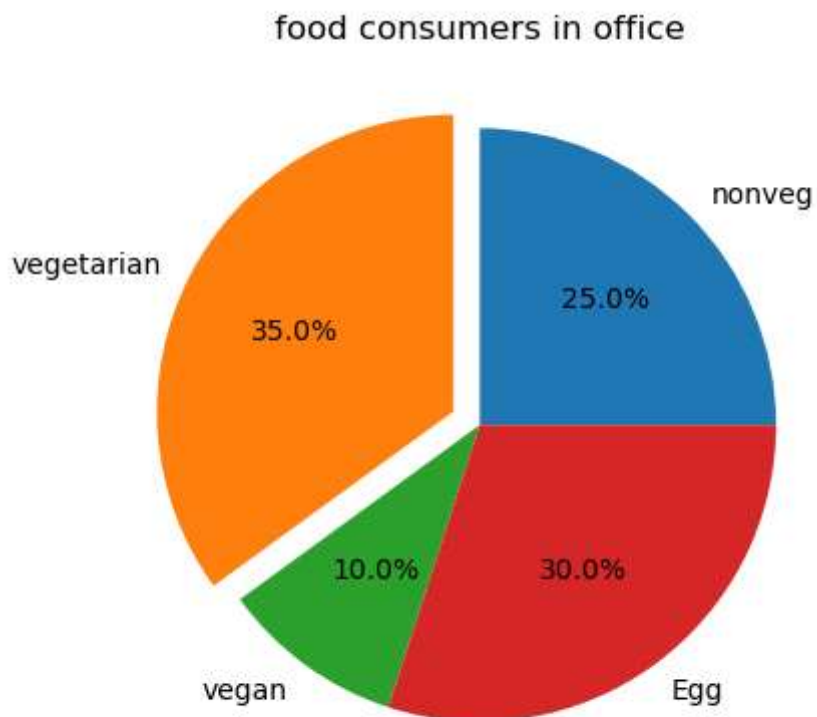
```
In [23]: values=[25,35,10,30]
label=["nonveg","vegetarian","vegan","Egg"]
plt.pie(values,labels=label,shadow=True)
plt.legend()
plt.title("food consumers in office")
plt.show()
```



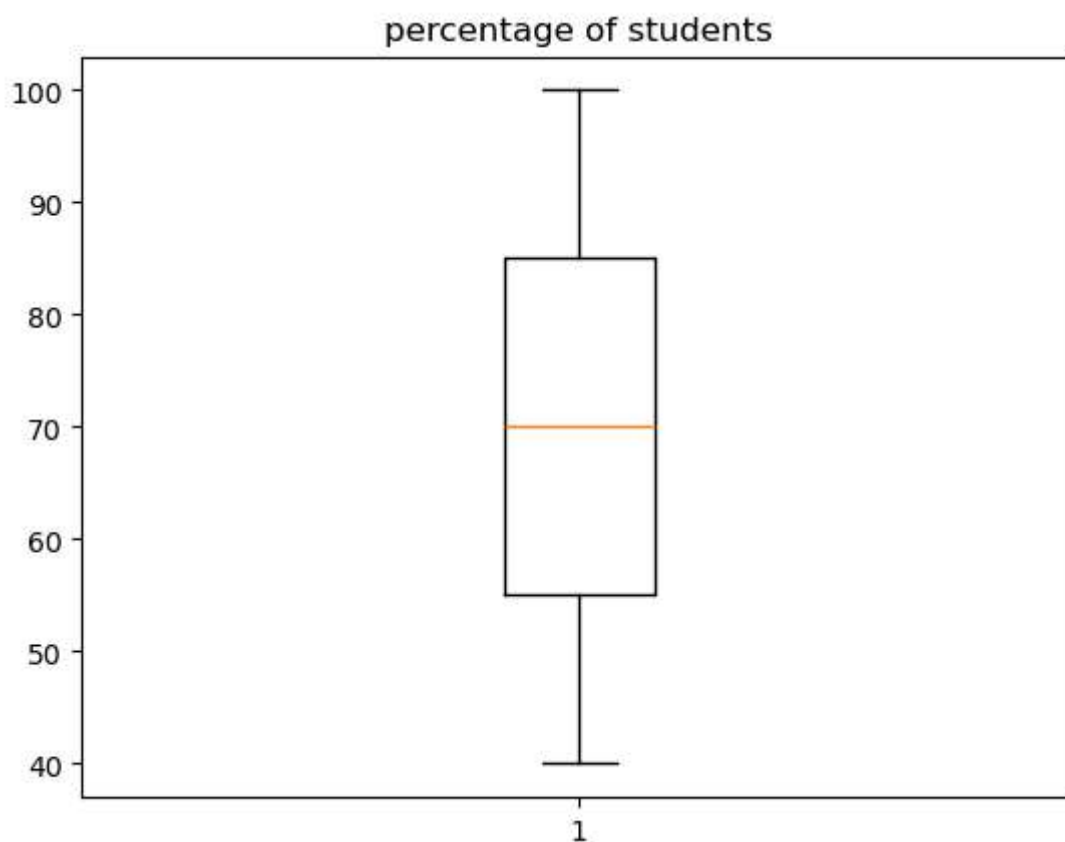
```
In [24]: values=[25,35,10,30]
label=["nonveg","vegetarian","vegan","Egg"]
plt.pie(values,labels=label,explode=(0,0.1,0,0),radius=1.5)
plt.title("food consumers in office")
plt.show()
```



```
In [25]: values=[25,35,10,30]
label=["nonveg","vegetarian","vegan","Egg"]
plt.pie(values,labels=label,explode=(0,0.1,0,0),radius=1,autopct='%1.1f%%')
plt.title("food consumers in office")
plt.show()
```



```
In [28]: percent_marks=np.array([40,50,60,70,80,90,100])
plt.title("percentage of students")
plt.boxplot(percent_marks)
plt.show()
```



minimum-smallest value-40

first quartile(Q1)-25 percentile-55

median(Q2)-50th percentile-70

third quartile(Q3)-85

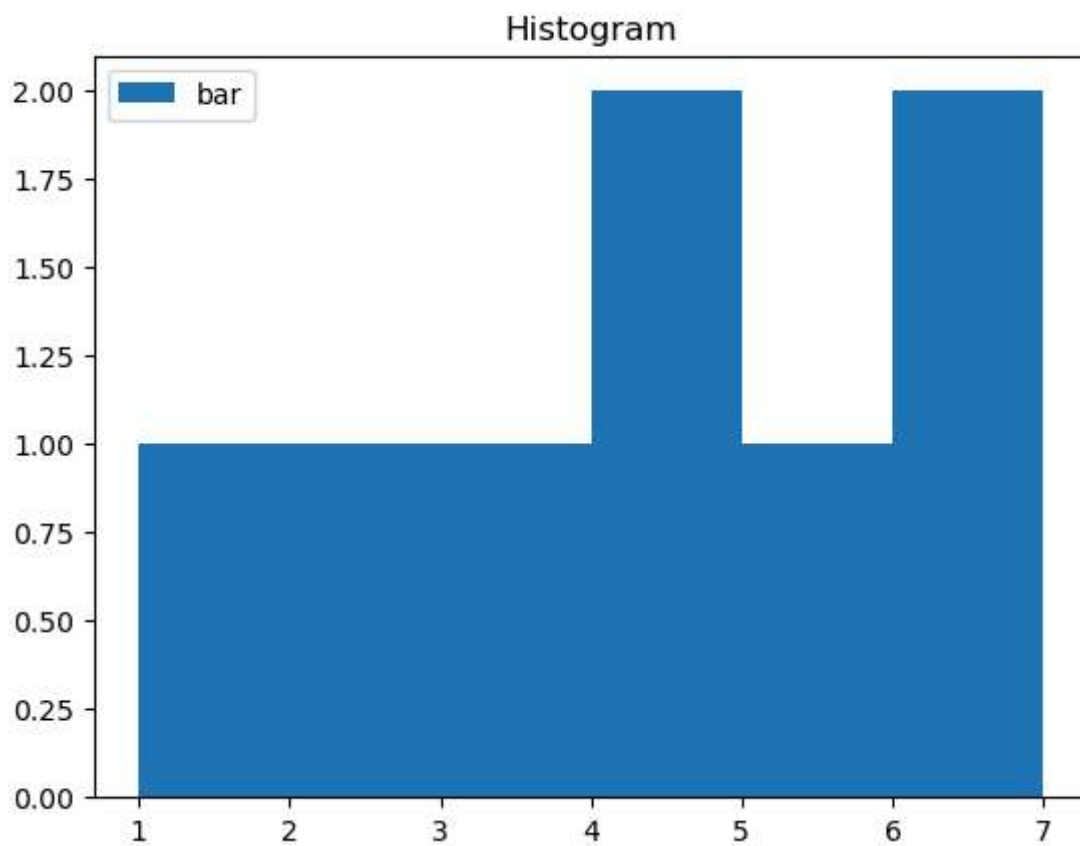
maximun-(Q4)-100

```
In [1]: import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5, 6, 7, 4]

plt.hist(x, bins = [1, 2, 3, 4, 5, 6, 7])

plt.title("Histogram")
plt.legend(["bar"])

plt.show()
```



```
In [ ]:
```